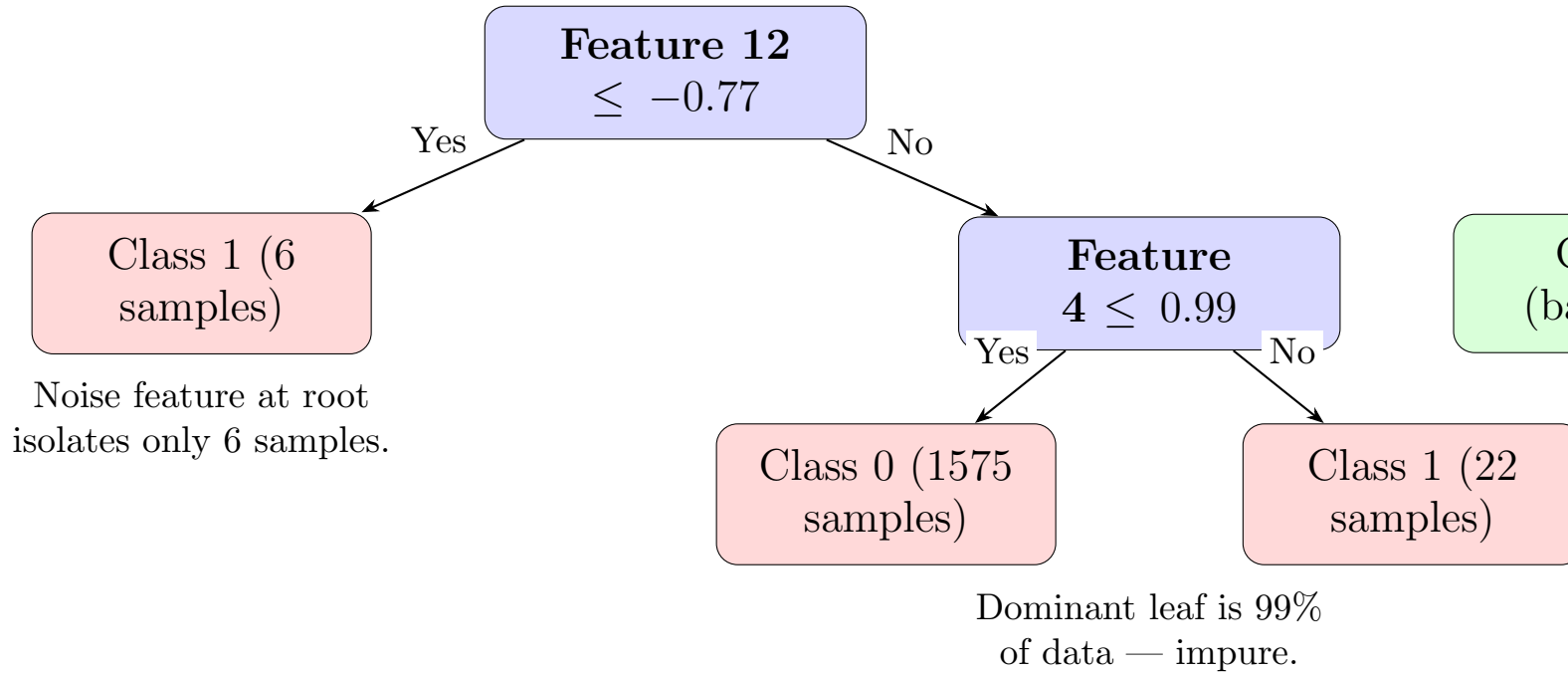
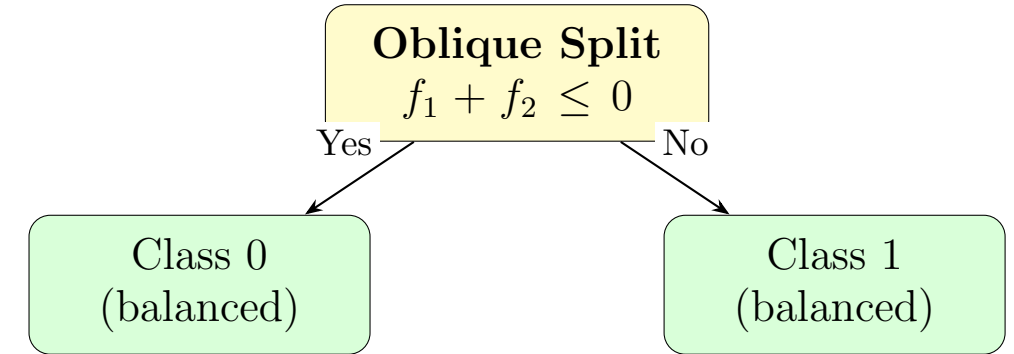


## Greedy C4.5 Tree (Suboptimal)



**Accuracy  $\approx 52\%$**

## Globally Optimal (Conceptual)



Requires look-ahead or oblique (non-axis-aligned) splits.  
C4.5 cannot do this greedily.

**Accuracy  $\approx 100\%$**

**Key Insight:** In XOR/parity problems, any single axis-aligned split is uninformative because both halves contain roughly equal numbers of both classes. Greedy splitting commits to a locally optimal noise feature at the root, leaving the vast majority of data in an impure leaf. A globally optimal tree would require evaluating split combinations or using oblique splits — both intractable for standard C4.5.